Homework 4

ASEN 5335 Aerospace Environment

Justin Le
October 18, 2026

Problem 1: Field Magnitudes

Using the WMM model from the NOAA calculator, we find the magnetic field in Boulder, CO to be,

Model Used:	WMM-2025						
Latitude:	40° 0' 59" N						
Longitude:	105° 17' 10" W						
Elevation:	5430.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	7° 40' 8"	66° 0' 55"	20,760.4 nT	20,574.7 nT	2,770.4 nT	46,662.2 nT	51,072.1 nT
Change/year	-0° 4' 50"/yr	-0° 2' 59"/yr	-8.9 nT/yr	-5.0 nT/yr	-30.2 nT/yr	-129.4 nT/yr	-121.9 nT/yr
Uncertainty	0° 22'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 1: Boulder Magnetic Field

We repeat with the same model for McMurdo Station Anarctica to find,

Model Used:	WMM-2025						
Latitude:	77° 50' 46" S						
Longitude:	166° 40' 5" E						
Elevation:	33.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	140° 5' 50"	-80° 5' 50"	10,647.5 nT	-8,168.0 nT	6,830.2 nT	-60,990.6 nT	61,913.0 nT
Change/year	-0° 12' 10"/yr	0° 2' 29"/yr	33.8 nT/yr	-1.7 nT/yr	50.6 nT/yr	66.8 nT/yr	-60.0 nT/yr
Uncertainty	0° 34'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 2: McMurdo Magnetic Field

TODO: Based on the inclination observed at McMurdo, this tells us that...

Problem 2: Dipole Field

(i).

Assuming a dipole field model with $L = 4$ $B_0 = 31,000$ nT, $R_E = 6371$ km, and at the equator ($\lambda = 0°$), we can use the following equations to calculate the magnetic field magnitude,

$$r = LR_E \cos^2 \lambda$$
$$|B| = B_0(\frac{R_E}{r})^3\sqrt{1 + 3\sin^2 \lambda}$$

$$\therefore r = 25,484,000$$
$$\boxed{\therefore |B| = 484.4 \text{ nT}}$$

(ii).

Repeating the above equations only for $\lambda = 30°$,

$$\therefore r = 19,113,000$$
$$\boxed{\therefore |B| = 1518.9 \text{ nT}}$$

Problem 3: Magnetopause

(i).

To find the standoff distance of the Earth to the magnetosphere, we can use the equation,

$$\rho_{sw} v_{sw}^2 = \frac{(2B_{mag})^2}{2\mu_0} = \frac{4B_0^2}{2\mu_0 L^6}$$
$$B_{mag} = B_0 \left(\frac{R_E}{r}\right)^3 = \frac{B_0}{L^3}, \qquad B_0 \approx 31{,}000 \text{ nT}$$
$$L = \left(\frac{2B_0^2}{\mu_0 \rho_{sw} v_{sw}^2}\right)^{1/6}$$

Given $v_{sw} = 400$ km/s, and $\mu_0 = 4\pi \times 10^{-7}$, and $\rho_{sw} = n m_p = 5$ proton/cm^3 $*1.672 \times 10^{-27}$ kg $= 8.36 \times 10^{-21}$ kg/m^3.

$$\boxed{L = 10.2}$$

(ii).

We can repeat the calculation above but instead using $v = 1000$ km/s, and $n = 10$ proton/cm^3, for a new $\rho_{sw} = 1.67 \times 10^{-20}$ kg/m^3.

$$\boxed{L = 6.7}$$

Problem 4: The Ring Current

(i).

(ii).

(iii).

Problem 5: Drift due to Gravity

Problem 6: Geomagnetic Indices

Problem 7: Geomagnetic Storm

Problem 8: Key Parameters in the Solar Wind Driving a Magnetic Storm

Problem 9: The Core

Problem 10: Magnetospheric Multiscale